PieClam

Inclusive Exclusive Cluster Affiliation Model With Prior

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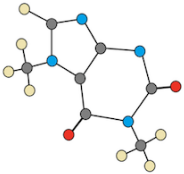
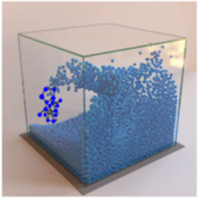
Machine Learning on Graphs



chemistry

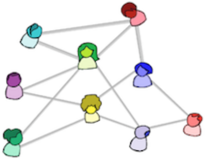
recommender systems

traffic prediction



simulation

surfaces



social networks

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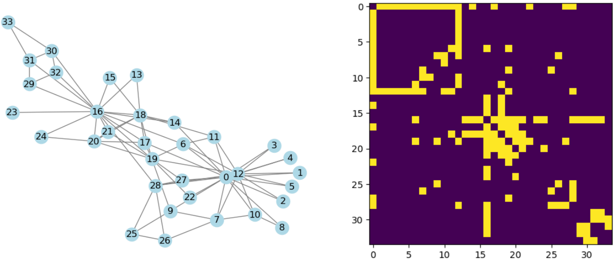
Notations

- A **graph** is a pair $G = ([N], E)$.
- $E \subseteq [N] \times [N]$ is the set of **edges**.
- A pair $(n, m) \in [N] \times [N]$ is called a **dyad**.
- **Sparse graphs**: $O(|E|) \lll N^2$
- An edge between nodes n and m is denoted by $n \sim m$. A non-edge by $n \nsim m$.
- $\mathbf{A} = \{a_{nm} \in \{0,1\}\}_{n,m=1}^{N}$ is the **adjacency matrix**.
- **Random Graph Model**: An $N \times N$ matrix with elements $p_{nm} = P(n \sim m)$
- **Undirected**: $n \sim m \iff m \sim n$

Notations



Zachari's karate club graph and it's adjacency matrix.

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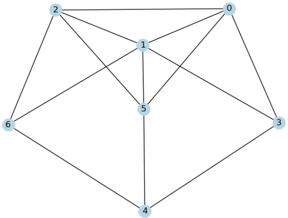
Graph Representation Learning

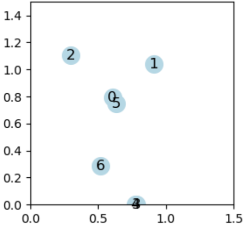
- Learn a vector $\mathbf{f}_n \in \mathbb{R}^C$ for every node, $C \lll N$.
  $\mathbf{F} \in \mathbb{R}^{N \times C}$ is the matrix whose rows are the features.

- The representation captures the **structural** information.

- Representation is used for downstream tasks:
  Examples: Community Detection, Node/Graph Classification, Anomaly Detection, Link Prediction, etc.



$\longrightarrow$



How to learn the embedding?

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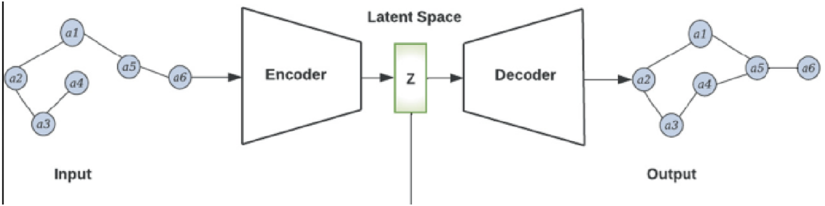
Autoencoders

- **Intuition**: A reconstructable representation captures the structural information.
- **Encoder f** : $[N] \to \mathbb{R}^{N \times C}$
- **Decoder** $w : \mathbb{R}^C \times \mathbb{R}^C \to [0,1]$. $w(\mathbf{f}_n, \mathbf{f}_m) = P(n \sim m)$
- Optimize by making $w(\mathbf{f}_n, \mathbf{f}_m)$ similar to $\mathbf{A}$.



Autoencoders: Factorization and Inner Product Decoders

- **Factorization** based decoders: $w(\mathbf{f}_n, \mathbf{f}_m) = g(\mathbf{f}_n^\top \mathbf{B} \mathbf{f}_m) \approx a_{nm}$.
- **Inner product** methods assume $w(\mathbf{f}_n, \mathbf{f}_m) = g(\mathbf{f}_n^\top \mathbf{f}_m) \approx a_{nm}$.
- Example: **BigClam** is an inner product method in which $w(\mathbf{f}_n, \mathbf{f}_m) = 1 - e^{-\mathbf{f}_n^\top \mathbf{f}_m} \approx a_{nm}$
- Similarly, proximity based reconstruction learns $w(\|\mathbf{f}_n - \mathbf{f}_m\|_2) \approx a_{nm}$ (Laplacian eigenmaps).

Message Passing

Message passing on graphs updates the representation of nodes with aggregated data from neighbors.

$$\mathbf{m}_n^t = \underset{k \in \mathcal{N}(n)}{Aggr} M^t(\mathbf{f}_n^t, \mathbf{f}_k^t)$$

$$\mathbf{f}_n^{t+1} = U^t(\mathbf{m}_n^t, \mathbf{f}_n^t)$$

$$R^t = \underset{n \in [N]}{Bggr}(\mathbf{F}^t)$$

Aggr and *Bggr* are permutation invariant (aggregation functions).

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Main Contributions

- **IeClam & PieClam**: novel factorization autoencoder methods.
- IeClam is based off of BigClam which is an effective and efficient inner product autoencoder model.
- We define a distance measure between probabilistic graph models called **log cut** distance, and a concept of **universality** for graph autoencoders.
- **Universality Theorem** for IeClam, a desired precision ϵ requires $C(\epsilon) = O(\epsilon^{-2})$ embedding dimension for any graph.
- Previous methods similar to BigClam **not universal**.
- Extend the embedding space to a probability space - PieClam and PClam are **generative** versions of IeClam and BigClam.

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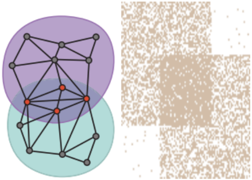
BigClam: Community Affiliation

- **Community**: a group of nodes with the same structural behavior.
- **Overlapping Communities**: same as autoencoded feature embeddings.
- **Inclusive communities**: common membership increases connection probability.
- Inner product decoding can *only* embed inclusive communities.



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BigClam: Overview

- **BigClam Overview:**
  - Inner product based (inclusive) overlapping community affiliation model.
  - $f_n^c \in [0, \infty)$ represents the affiliation of node n to community c.
  - Proven very efficient and effective.

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BigClam: Probabilistic Model

- Non edges are Poisson distributed with mean $\mathbf{f}_n^\top \mathbf{f}_m$ $(k = 0)$.

$$P(n \nsim m | \mathbf{f}_n, \mathbf{f}_m) = e^{-\mathbf{f}_n^\top \mathbf{f}_m}$$

- Therefore

$$P(n \sim m | \mathbf{f}_n, \mathbf{f}_m) = 1 - e^{-\mathbf{f}_n^\top \mathbf{f}_m}$$

- The probability for the entire graph is

$$P(E|\mathbf{F}) = \sqrt{\prod_{n \in [N]} \prod_{m \in \mathcal{N}(n)} P(n \sim m | \mathbf{f_n}, \mathbf{f_m}) \prod_{m \notin \mathcal{N}(n)} P(n \nsim m) | \mathbf{f}_n, \mathbf{f}_m)}$$

- The log likelihood is

$$l(\mathbf{F}) = \frac{1}{2}\sum_{n\in[N]}\Bigg(\sum_{m\in\mathcal{N}(n)}\log(1-e^{-\mathbf{f}_n^\top\mathbf{f}_m}) - \sum_{m\notin\mathcal{N}(n)}\mathbf{f}_n^\top\mathbf{f}_m\Bigg).$$

- Optimize by gradient ascent:

$$\mathbf{F}^{(i+1)} = \mathbf{F}^{(i)} + \delta\nabla_{\mathbf{F}} l(\mathbf{F}^{(i)}), \qquad (1)$$

for some learning rate $\delta > 0$.

BigClam: Reparametrization Trick

- Reparametrization trick

$$2l(\mathbf{F}) = \sum_{n\in[N]}\Bigg(\sum_{m\in\mathcal{N}(n)} \log\left(e^{\mathbf{f}_n^\top \mathbf{f}_n} - 1\right) - \mathbf{f}_n^\top \sum_{n\in[N]} \mathbf{f}_m + \|\mathbf{f}_n\|^2\Bigg).$$

- The gradient

$$\nabla_{\mathbf{f}_n} l = \sum_{m\in\mathcal{N}(n)} \mathbf{f}_m \left(1 - e^{-\mathbf{f}_n^\top \mathbf{f}_m}\right)^{-1} - \sum_{n\in[N]} \mathbf{f}_m + \mathbf{f}_n.$$

The number of operations is now $O(|E|)$, every node is updated by the same aggregation of it's neighborhood (**message passing**).

BigClam as Message Passing

The feature optimization is implemented as a message passing algorithm.

$$
\begin{aligned}
M^t(\mathbf{f}_n^t, \mathbf{f}_m^t) &= \mathbf{f}_m^t \left(1 - e^{-\mathbf{f}_n^{t\top} \mathbf{L} \mathbf{f}_m^t}\right)^{-1} \\
\mathbf{m}_n^t &= \sum_{k \in \mathcal{N}(n)} M^t(\mathbf{f}_n^t, \mathbf{f}_k^t) \\
R^t &= \sum_{n \in [N]} \mathbf{f}_n^t \\
\mathbf{f}_n^{t+1} &= U(\mathbf{m}_n^t, \mathbf{f}_n^t, \delta) = \mathbf{f}_n^t + \delta(\mathbf{m}_n^t - R^t + \mathbf{f}_n^t)
\end{aligned}
\tag{2}
$$

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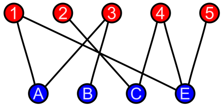
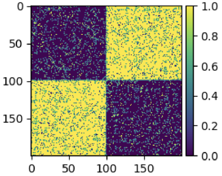
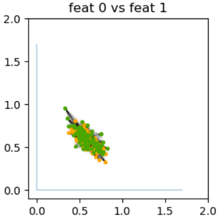
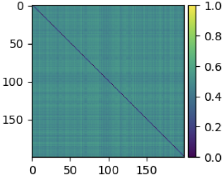
IeClam: Bipartite is not Inclusive





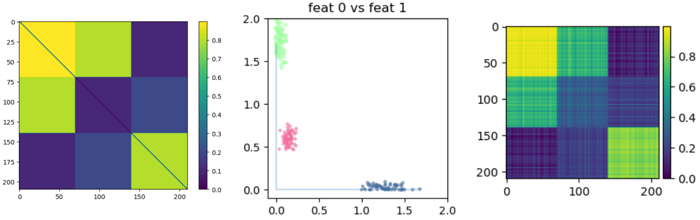
Triangle inequality: if $n \sim k$ and $m \sim k$ then $\mathbf{f}_n^\top \mathbf{f}_k$ and $\mathbf{f}_m^\top \mathbf{f}_k$ are large and therefore $\mathbf{f}_n^\top \mathbf{f}_m$ is also large which implies $m \sim n$ with high probability.





Bipartite autoencoding with BigClam

Inclusive model decodes bipartite as one inclusive community.



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IeClam: Exclusive Communities

- **Exclusive community**: a community for which common membership *reduces* the probability to connect.
- Two nodes with a high score for the same exclusive community c should have a lower connection probability for it.
- Denote inclusive communities with $\mathbf{t} \in \mathbb{R}^C$ and exclusive communities with $\mathbf{s} \in \mathbb{R}^C$.

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IeClam: Probabilistic Model

- The representation for node n will be: $\mathbf{f}_n = (\mathbf{t}_n, \mathbf{s}_n)$.
- Instead of inner product use **L-Product**:

$$\mathbf{f}_n^\top \mathbf{L} \mathbf{f}_m = \mathbf{t}_n^\top \mathbf{t}_m - \mathbf{s}_n^\top \mathbf{s}_m$$

- The single edge probability is now:

$$P(n \sim m | \mathbf{f}_n, \mathbf{f}_m) = 1 - e^{-\mathbf{f}_n^\top \mathbf{L} \mathbf{f}_m}$$

Where $\mathbf{L} = \mathrm{diag}(1, \ldots, 1, -1, \ldots, -1)$

- **Notice**: for a probabilistic model $\mathbf{f}_n^\top \mathbf{L} \mathbf{f}_m \geq 0 \iff \mathbf{t}_n^\top \mathbf{t}_m \geq \mathbf{s}_n^\top \mathbf{s}_m$.

IeClam: Probabilistic Model

- Inner product has been replaced by L product.
- The log likelihood

$$l(\mathbf{F}) = \frac{1}{2}\sum_{n\in[N]}\Big(\sum_{m\in\mathcal{N}(n)}\log(1-e^{-\mathbf{f}_n^\top\mathbf{L}\mathbf{f}_m}) - \sum_{m\notin\mathcal{N}(n)}\mathbf{f}_n^\top\mathbf{L}\mathbf{f}_m\Big).$$

- Reparametrization trick

$$2l(\mathbf{F}) = \sum_{n\in[N]}\Big(\sum_{m\in\mathcal{N}(n)}\log(e^{\mathbf{f}_n^\top\mathbf{L}\mathbf{f}_n}-1) - \mathbf{f}_n^\top\mathbf{L}\sum_{n\in[N]}\mathbf{f}_m + \mathbf{f}_n^\top\mathbf{L}\mathbf{f}_n\Big).$$

- The gradient

$$\nabla_{\mathbf{f}_n} l = \mathbf{L}\Bigg(\sum_{m\in\mathcal{N}(n)}\mathbf{f}_m\left(1-e^{-\mathbf{f}_n^\top\mathbf{L}\mathbf{f}_m}\right)^{-1} - \sum_{n\in[N]}\mathbf{f}_m + \mathbf{f}_n\Bigg).$$

- Complexity stays $O(|E|)$.

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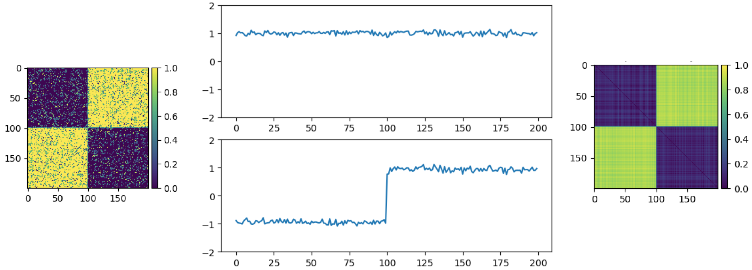
- A bipartite graph $\mathbf{A}$ can be encoded by embedding part 1 to (b, b) and part 2 to $(b, -b)$ where $b \in \mathbb{R}_+$.

$$w(\mathbf{f}_n, \mathbf{f}_m) = \begin{cases} 1 - e^{-b^2} & \text{if } m, n \textbf{ not} \text{ the same part} \\ 0 & \text{otherwise} \end{cases}$$

- $w(\mathbf{f}_n, \mathbf{f}_m) \to \mathbf{A} \quad \text{as} \quad b \to \infty.$



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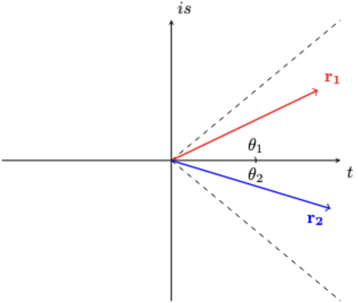
- The bilinear form $\mathbf{L}$ is **not** an inner product.
- **Positive probabilities**: restrict the community affiliation space to $\mathcal{C} \subset \mathbb{R}^{2C}$ s.t $\forall \mathbf{f}, \mathbf{g} : \mathbf{f}^{\top}\mathbf{L}\mathbf{g} > 0$.
- For simplicity, we further restrict $\mathbb{R}^{2C}$ to pairwise cones

$$\mathcal{T} = \{\mathbf{f} = (\mathbf{t}, \mathbf{s}) s.t. \forall c \in [C] : |s^c| \leq t^c\}$$

- $\implies$ The most exclusive you can get is on $s^c = \pm t^c$.

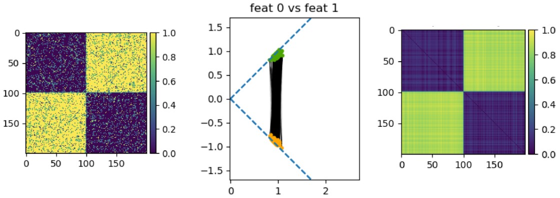
IeClam: Pairwise Cones - Intuition

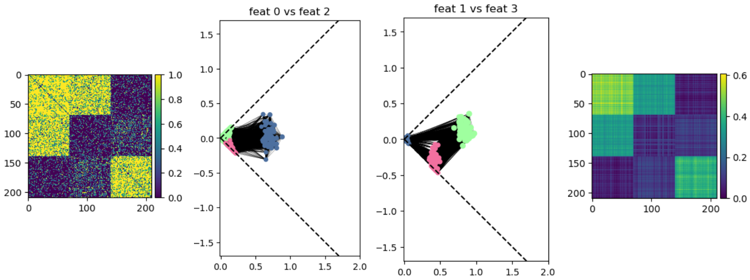
- We can think of (s^c, t^c) as one complex feature $f^c = t^c + is^c = r^c e^{i\theta^c}$.
- For two nodes: $\mathbf{f}_1^{c\top} \mathbf{L} \mathbf{f}_2^c = \mathcal{R}(f_1^c f_2^c) = r_1 r_2 \cos(\theta_1 + \theta_2)$



- $\implies$ The most exclusive you can get is on $s = \pm t$.

Pairwise Cones: Examples





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PieClam: Generative Models

- $P(E|\mathbf{F})$ restricted to the training graph.
- Learn the joint probability distribution $P(E \wedge \mathbf{F})$.
- **Motivation**:
  1. Generate new graphs by sampling.
  2. Infer the likelihood of other graphs.

PieClam: Prior on Code Space

- Define a prior probability density function code space.
- Using Bayes' law:

$$p(E \wedge \mathbf{F}) = P(E|\mathbf{F})p(\mathbf{F})$$

- Assume independence of feature events:

$$p(\mathbf{F}) = \prod_{n \in [N]} p(\mathbf{f}_n)$$

- The probabilistic model will now be:

$$p(E \wedge \mathbf{F}) = \underbrace{\prod_{n \in [N]} p(\mathbf{f}_n)}_{p(\mathbf{F})} \underbrace{\left(\sqrt{\prod_{m \in \mathcal{N}(n)} P(n \sim m|\mathbf{F}) \prod_{m \notin \mathcal{N}(n)} P(n \nsim m|\mathbf{F})} \right)}_{P(E|\mathbf{F})}$$

PieClam: Probabilistic Model and PClam

- log likelihood loss:

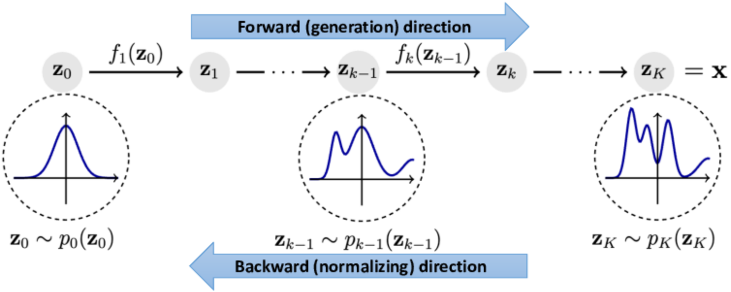
$$l(\mathbf{F}) = \sum_{n\in[N]} \left(\frac{1}{2} \Big(\sum_{m\in\mathcal{N}(n)} \log(e^{\mathbf{f}_n^\top \mathbf{L} \mathbf{f}_n} - 1) - \mathbf{f}_n^\top \mathbf{L} \sum_{n\in[N]} \mathbf{f}_m + \mathbf{f}_n^\top \mathbf{L} \mathbf{f}_n \Big) + \log\left(p(\mathbf{f}_n)\right) \right).$$

- The prior acts as **regularization** - attractive force.
- **PClam**: BigCalm with prior.
- Model the prior as a **neural network** $p(\mathbf{f}|\alpha)$.

- An invertible transformation $\mathbf{z}$ from the affiliation probability space to a gaussian probability space.
- The transformation is scaled so the distribution stays normalized.

$$p(\mathbf{f}) = G\Big(\mathbf{z}(\mathbf{f}) \cdot |\det(\partial_f \mathbf{z})|\Big)$$

- Maximize log likelihood of the features.



PieClam: Optimizing PieClam

- **Node initialization**: random positions according to gaussian in the middle of $\mathbb{R}_+^{\mathbb{C}}$ or $\mathcal{T}$.
- Run **alternating** optimization on the affiliation features $\mathbf{f}$ and prior parameters α.
- Optimize each of the two losses with **gradient ascent**.
- **Feature regularization**: l_1 regularization.
- **s regularization**: l_1 regularization on the **s** communities.
- **Prior regularization**: Noise addition and weight decay.
- **More hypers**: learning rate and number of iterations for feats and prior optimizations, number of alternations, learning rate scheduler.

PieClam as Graphon

- A graph generative model that extends SBMs.
- Can be seen as a weighted graph with a "continuous" node set $[0,1]$.

Definition

The space of graphons $\mathcal{W}$ is defined to be the set of all measurable functions $W : [0,1]^2 \to [0,1]$ which are symmetric, namely $W(x,y) = W(y,x)$.

- The edge weight $W(x,y)$ of a graphon $W \in \mathcal{W}$ can be seen as the probability of having an edge between node x and node y.
- A random graph is generated by sampling independent uniform nodes $\{X_n\}$ from the graphon domain $[0,1]$, and connecting each pair X_n, X_m in probability $W(X_n, X_m)$.

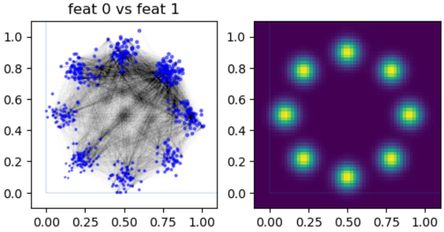
PieClam as Graphon

- There is a probability preserving a.e. bijection $\xi_p : [0,1] \to \mathbb{R}^K$ that maps the prior probability to the uniform probability over $[0,1]$ (another NF layer).
- Sample N points $\{X_n\}_{n=1}^N \subset [0,1]$ uniformly. Observe that $\{\xi_p(X_n)\}_{n=1}^N \subset \mathbb{R}^K$ are indepndent samples via the probability density p.
- Connect the points according to the BigClam or IeClam models $P(n \sim m | \xi_p(X_n), \xi_p(X_m))$.

  This shows that PClam and PieClam coincide with the generative graphon model.

Sample a graph with N nodes from PClam or PieClam,

- Sample affiliation vectors $(\mathbf{f}_n)_{n=0}^{N} \sim p(\cdot)$.
- Connect the nodes with probability $P(n \sim m | \mathbf{f}_n, \mathbf{f}_m) = 1 - e^{\mathbf{f}_n^\top L \mathbf{f}_m}$.



PieClam: Graphs With Node Features

- Definition: a **graph-signal** is the ordered triplet $([N], E, \mathbf{X})$, $\mathbf{X} \in \mathbb{R}^d$ are **node features** given with the data.
- Examples: user info, protein composition, social media groups etc.
- PieClam (and Pclam) for a graph-signal: concatenate $(\mathbf{f}_n, \mathbf{x}_n)$ when learning the prior.

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Universal Autoencoders

For some distance measure for $d(\cdot,\cdot)$

Definition (Universal Autoencoder)

A family of code spaces $\mathbb{R}^M$ and corresponding pairwise decoders $D_M : \mathbb{R}^{2M} \longrightarrow [0,1]$, parametrized by $M \in \mathbb{N}$, is called *universal* if for every $\epsilon > 0$ there is $M \in \mathbb{N}$ (which depends only on ϵ) such that for every $N \in \mathbb{N}$ and every graph with adjacency matrix $\mathbf{A}$ and N nodes there are N points in the code space $\{z_n \in \mathbb{R}^M\}_{n=1}^N$ such that

$$d(\mathbf{D}_M(\mathbf{z}), \mathbf{A}) < \epsilon.$$

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- The **cut norm** of a matrix $\mathbf{X} \in \mathbb{R}^{N\times N}$ is defined to be:

$$\|\mathbf{X}\|_\square := \frac{1}{N^2} \sup_{\mathcal{U},\mathcal{V}\subset[N]} \left| \sum_{i\in\mathcal{U}} \sum_{j\in\mathcal{V}} x_{i,j} \right|. \tag{3}$$

- The **cut metric** between matrices $\mathbf{A}$ and $\mathbf{B}$ is $\|\mathbf{A}-\mathbf{B}\|_\square$

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Universality: Log Cut Distance

- The **Log Cut Metric**:

$$D_{\square}(\mathbf{P}||\mathbf{Q}) := \frac{1}{N^2} \sup_{\mathcal{U},\mathcal{V}\subset[N]} \left(\left| \log \left(\prod_{n\in\mathcal{U}} \prod_{m\in\mathcal{V}} \frac{1-p_{n,m}}{1-q_{n,m}} \right) \right| \right)$$

$$= \| \log(1-\mathbf{P}) - \log(1-\mathbf{Q}) \|_{\square}$$

Interpretation: The biggest part of the model P that can't be explained by the model Q.

Universality - Log cut Distance

- If one of the matrices has elements 1 (e.g. a **realization A**), regularization is added and penalized: **P** and **A** is defined to be

$$D_{\square}(\mathbf{P}||\mathbf{A}) := \inf_{0<d\leq 1}\left(d + \frac{1}{N^2}\sup_{\mathcal{U},\mathcal{V}\subset[N]}\left|\log\left(\prod_{n\in\mathcal{U}}\prod_{m\in\mathcal{V}}\frac{1-p_{n,m}}{1-(1-d)a_{n,m}}\right)\right|\right).$$

- This version of the log cut is used in the universality theorem.

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Universal Autoencoders with Log Cut

Definition (Universal Autoencoder)

A family of code spaces $\mathbb{R}^M$ and corresponding pairwise decoders $D_M : \mathbb{R}^{2M} \longrightarrow [0,1]$, parametrized by $M \in \mathbb{N}$, is called *universal* if for every $\epsilon > 0$ there is $M \in \mathbb{N}$ (which depends only on ϵ) such that for every $N \in \mathbb{N}$ and every graph with adjacency matrix $\mathbf{A}$ and N nodes there are N points in the code space $\{z_n \in \mathbb{R}^M\}_{n=1}^N$ such that

$$D_\square(\mathbf{D}_M(\mathbf{z})||\mathbf{A}) < \epsilon.$$

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Universality in $\mathbb{R}^{2K}$

We prove that IeClam (and consequently PieClam) is universal.

Theorem

For every epsilon $\epsilon > 0$, every $N \in \mathbb{N}$, and every adjacency matrix $\mathbf{A} \in [0,1]^{N \times N}$, there are N affiliation features $\mathbf{F} \in \mathbb{R}^{2K}$ of dimension $K = -9\frac{\log(\frac{\epsilon}{2})^2}{\epsilon^2}$ such that the corresponding IeClam model $\mathbf{P} = \{P(n \sim m|\mathbf{f}_n, \mathbf{f}_m)\}_{n,m=1}^{N}$ satisfies

$$D_{\square}(\mathbf{P}||\mathbf{A}) < \epsilon.$$

As a result, IeClam and PieClam are universal autoencoders with code space $\mathbb{R}^{2K}$.

Universality in the unrestricted space requires $O(\epsilon^-2)$ communities.

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Universality in $\mathcal{T}$

Theorem

For every epsilon $\epsilon > 0$, every $N \in \mathbb{N}$, and every adjacency matrix $\mathbf{A} \in [0,1]^{N\times N}$, there are N affiliation features $\mathbf{F}$ in the cone of pairwise non-negativity $\mathcal{T} \subset \mathbb{R}^{2C}$ of dimension $C = 2^{4\lceil -\log(\epsilon/2)^2/\epsilon^2\rceil}$ such that the corresponding IeClam model $\mathbf{P} = \{P(n \sim m|\mathbf{f}_n, \mathbf{f}_m)\}_{n,m=1}^{N}$ satisfies IeClam and PieClam are universal autoencoders with code space $\mathcal{T}$.

Universality in positivity cones is $O(2^{\frac{1}{\epsilon^2}})$ (unrealistically huge!)

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BigClam and PClam Are Not Universal

- For BigClam and PClam we show that the dimension of the embedding space will have to **depend on the number of nodes** in the graph.
- The proof is similar to what was shown in a previous slide.

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Experiments

- **Experiment Overview:**
  - Synthetic prior reconstruction.
  - Synthetic SBM approximation with distance metrics.
  - Anomaly detection.
- **Key Results:**
  - synthetic priors in two dimensions are visibly reconstructed only from the graph structure.
  - Off diagonal SBMs are decoded well with Ieclam and PieClam and badly by BigClam and PClam.
  - PieClam and IeClam are competative as an anomaly detectors.
  - PieClam is compentative as an anomaly detector on graph-signals without using the node features.

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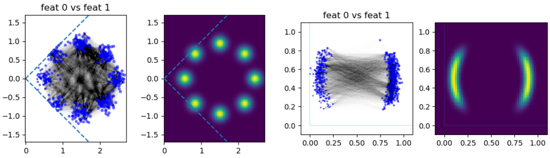
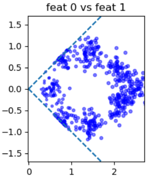
Experiments: Synthetic Priors - Description

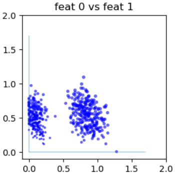
- Sample nodes from synthetic priors and try to learn the prior.
- Resample the affiliation vectors from a standard gaussian in the middle of the code space.
- We optimize BigClam, PClam, IeClam and PieClam to reconstruct the shape they were sampled from, and learn the prior.

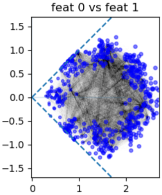


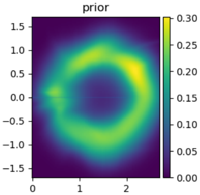
Experiments: Synthetic Priors - Results

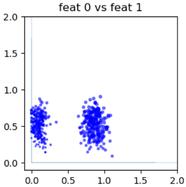
- Priors have been reconstructed to a similar degree.
- Shape also reconstructed with BigClam and IeClam.
- Prior regularizes results.

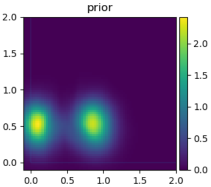












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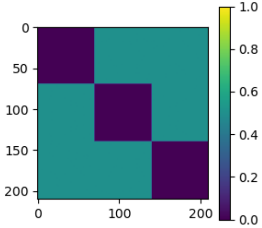
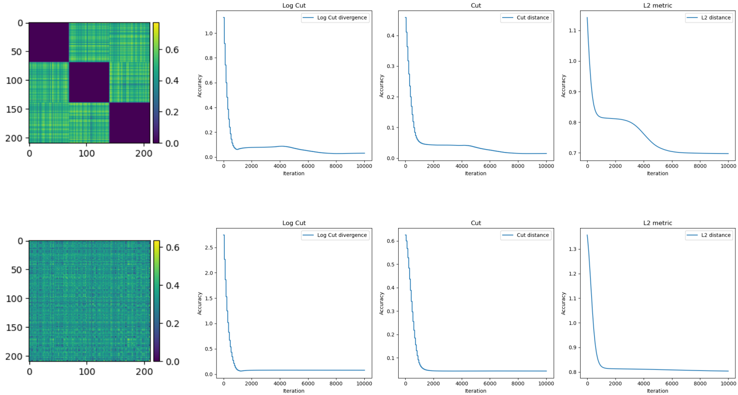
Experiments: SBM Distance Measurement - Description

- We sample graphs from the SBMs shown below.
- We approximate the SBMs with the four models and measure the cut, log cut and l2 distances.



Experiments: SBM Distance Measurement - Results

- Inclusive methods can't learn exclusive structures.
- Inclusive Exclusive methods reach lower losses.
- Exclusive methods find the SBMs well.



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Experiments: Anomaly Detection - Description

- Calculate node probability and classify with AUC score.
- Probability calculation:
  - star probability **(S)**
  - Prior probability **(P)**
  - (P) + (S) **(PS)**
- Compare to state of the art, datasets with real anomalies: reddit, elliptic, photo.

Using node features.

Method	Reddit	Elliptic	Photo
(S)- IeClam	**0.639**	0.440	**0.592**
(S) - PieClam	0.6320 $\pm$ 0.0054	0.4354 $\pm$ 0.0005	*0.5727 $\pm$ 0.0151
(P) - PieClam	0.5089 $\pm$ 0.0219	**0.6201 $\pm$ 0.0176**	0.4252 $\pm$ 0.0074
(PS) - PieClam	*0.6329 $\pm$ 0.0052	<u>0.5294 $\pm$ 0.0089</u>	0.5289 $\pm$ 0.0127
(S) - BigClam	<u>0.637</u>	0.434	<u>0.581</u>
DOMINANT	0.511	0.296	0.514
AnomalyDAE	0.509	*0.496	0.507
OCGNN	0.525	0.258	0.531
AEGIS	0.535	0.455	0.552
GAAN	0.522	0.259	0.430
TAM	0.606	0.404	0.568

Table: First place in **boldface**, second with <u>underline</u>, third with *star.

Without node features - Still competative!

Method	Reddit	Elliptic	Photo
(S) - PieClam	0.6320 ± 0.0054	0.4354 ± 0.0005	$\mathbf{0.5727 \pm 0.0151}$
(P) - PieClam	0.5089 ± 0.0219	$\mathbf{0.6201 \pm 0.0176}$	0.4252 ± 0.0074
(PS) - PieClam	*0.6329 ± 0.0052	$\underline{0.5294 \pm 0.0089}$	0.5289 ± 0.0127
(S) - PieClam (No Attr)	$\underline{0.6346 \pm 0.0015}$	0.4349 ± 0.0009	*0.5696 ± 0.0065
(P) - PieClam (No Attr)	0.4530 ± 0.0142	*0.4826 ± 0.0134	0.4977 ± 0.0747
(PS) - PieClam (No Attr)	$\mathbf{0.6351 \pm 0.0015}$	0.4349 ± 0.0009	$\underline{0.5697 \pm 0.0065}$

Table: Comparison of attributed and unattributed PieClam anomaly detection.

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Conclusion

- **Summary:**
  - IeClam extends BigClam to include both inclusive and exclusive communities.
  - PieClam and PClam extend IeClam and BigClam to generative models.
  - Proven universality in autoencoding any graph.
  - Effective in graph anomaly detection, synthetic reconstruction.
- **Future Work:**
  - Better Extension of PieClam to incorporate node features in edge probabilities.
  - Modifications for very sparse datasets.
  - Link prediction, node classification, etc.

Conclusion

Questions?

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BigClam Is Not Universal

- Consider the bipartite graph $\mathbf{B}$ with N nodes at each part, and probability $1 - e^{-a^2}$ for an edge between the two parts, and 0 within each part.
- Let $\mathbf{P} = 1 - e^{-\mathbf{F}^\top \mathbf{F}}$ be a decoded BigClam graph from the affiliation features $\mathbf{F}$.
- Denote $\tilde{\mathbf{P}}$ and $\tilde{\mathbf{B}}$ by

$$\tilde{p}_{n,m} = -\log(1 - p_{n,m}) = \mathbf{f}_n^\top \mathbf{f}_m,$$
$$\tilde{b}_{n,m} = -\log(1 - b_{n,m})$$

Claim

Under the above construction,

$$D_\square^0(\mathbf{P}||\mathbf{B}) \geq \frac{a^2}{16}.$$

BigClam Is Not Universal

Proof

Note that $\tilde{b}_{n,m} = a^2$ if n, m are in opposite parts, and $\tilde{b}_{n,m} = 0$ if n, m are on the same side. We index the nodes such that $[N]$ is the first side of the graph, and $[N] + N$ is the second side. For $n \in [N]$ we denote $\mathbf{q}_n = \mathbf{f}_n$ and $\mathbf{y}_n = \mathbf{f}_{n+N}$. Next, we use the identity

$$D^0_{\square}(\mathbf{P}||\mathbf{Q}) = \|\tilde{P} - \tilde{B}\|_{\square},$$

and bound the right-hand-side from below.

BigClam Is Not Universal

First, by the definition of cut norm, for every $\mathcal{U},\mathcal{V} \subset [2N]$,

$$\|\tilde{P}-\tilde{B}\|_{\square} \geq \left|\frac{1}{4N^2}\sum_{n\in\mathcal{U}}\sum_{m\in\mathcal{V}}(\tilde{p}_{n,m}-\tilde{b}_{n,m})\right|.$$

Hence, for $\mathcal{U}_1=\mathcal{V}_1=[N]$, $\mathcal{U}_2=\mathcal{V}_2=[N]+N$, $\mathcal{U}_3=[N],\mathcal{V}_3=[N]+N$, and $\mathcal{U}_4=[N]+N,\mathcal{V}_4=[N]$, we have

$$\|\tilde{P}-\tilde{B}\|_{\square} \geq$$

$$\frac{1}{16N^2}\sum_{j=1}^{4}\left|\sum_{n\in\mathcal{U}_j}\sum_{n\in\mathcal{V}_j}(\tilde{p}_{n,m}-\tilde{b}_{n,m})\right|$$

$$= \frac{1}{16N^2}\sum_{n=1}^{N}\sum_{m=1}^{N}\mathbf{q}_n^\top\mathbf{q}_m + \frac{1}{16N^2}\sum_{n=1}^{N}\sum_{m=1}^{N}\mathbf{y}_n^\top\mathbf{y}_m$$

$$+\frac{1}{8N^2}\sum_{n=1}^{N}\sum_{m=1}^{N}(a^2-\mathbf{q}_n^\top\mathbf{y}_m). \tag{4}$$

BigClam Is Not Universal

Denote

$$\mathbf{q} = \frac{1}{N}\sum_{n=1}^{N}\mathbf{q}_n, \quad \mathbf{y} = \frac{1}{N}\sum_{n=1}^{N}\mathbf{y}_n.$$

With these notations, (4) can be written as

$$\|\tilde{P} - \tilde{B}\|_\square \geq$$

$$\frac{1}{16}\left(\mathbf{q}^\top\mathbf{q} + \mathbf{y}^\top\mathbf{y} + 2\left|a^2 - \mathbf{q}^\top\mathbf{y}\right|\right) =$$

$$\begin{cases} \frac{1}{16}\left(a^2 + (\mathbf{q}^\top - \mathbf{y}^\top)(\mathbf{q} - \mathbf{y})\right) \geq \frac{a^2}{16} & \text{if} \quad a^2 > \mathbf{q}^\top\mathbf{y} \\ \frac{1}{16}\left(a^2 + (\mathbf{q}^\top - \mathbf{y}^\top)(\mathbf{q} - \mathbf{y})\right) \geq \frac{a^2}{8} & \text{otherwise} \end{cases}$$

In either case, the claim is proven.

□

BigClam With No Self Loops Approximating Bipartite Graphs

If we redefine the BigClam decoder to have no self-loops, namely, $\mathbf{P}$ has entries $p_{n,m} = P(n \sim m|\mathbf{f}_n, \mathbf{f}_m)$ for $n \neq m$, and $p_{n,m} = 0$ for $n = m$, then one can obtain a bipartite $\mathbf{P}$ with $C = N^2$ communities as follows.
Encode each node $n \in [N]$ in part 1 to $\mathbf{f}_n$ with $f_n^c = a$ for $c \in (n-1)N + [N]$ and $f_n^c = 0$ otherwise. Encode every node $n \in [N] + N$ from side 2 to $\mathbf{f}_n$ with $f_n^c = a$ for $c \in ([N]-1)N + n$ and $f_n^c = 0$ otherwise.
It is easy to see that the corresponding $\mathbf{P}$ is bipartite with edge probability between the parts being $1 - e^{-a^2}$.

PieClam Is Universal

The proof is based on a version of the weak regularity lemma for intersecting communities.

Definition

A (hard) intersecting community graph (ICG) with N nodes and K communities is a matrix $\mathbf{C} \in \mathbb{R}^{N \times N}$ of the following form. There exist K column vectors $\mathbf{Q} = (\mathbf{q}_k \in \{0,1\}^N))_{k=1}^K \in \{0,1\}^{N \times K}$ and k coefficients $\mathbf{r} = (r_k \in \mathbb{R})_{k=1}^K \in \mathbb{R}^K$ such that

$$\mathbf{C} = \mathbf{Q}\mathrm{diag}(\mathbf{r})\mathbf{Q}^\top.$$

Theorem

Let $\mathbf{A} \in \{0, R\}^{N \times N}$ be an adjacency matrix of a graph with N nodes. Let $\epsilon > 0$. Denote $K = \frac{9R^2}{4\epsilon^2}$. Then, there exists a hard ICG $\mathbf{C}$ with K communities such that

$$\|\mathbf{A} - \mathbf{C}\|_\square \leq \epsilon. \tag{5}$$

PieClam Is Universal

Let $\epsilon > 0$. Let $\mathbf{A} \in [0,1]^{N\times N}$ be an adjacency matrix.

We want to find an IeClam matrix $\mathbf{P} = 1 - \exp(-\mathbf{f}_n^\top \mathbf{L}\mathbf{f}_m)$ s.t.

$$D(\mathbf{P}\|\mathbf{A}) \le \epsilon$$

Define: $\tilde{\mathbf{A}} = \Big(-\log(1-(1-\epsilon)a_{n,m})\Big)_{n,m\in[N]\times[N]}$

PieClam Is Universal

Notice $\tilde{\mathbf{A}}$ has only two values

$$\tilde{a}_{nm} = \begin{cases} 0 & \text{if } a_{nm} = 0 \\ -\log(\epsilon) & \text{if } a_{nm} = 1 \end{cases}$$

$\implies$ We can use the weak regularity lemma:

There exists an ICG $\mathbf{C}$ for which

$$\|\tilde{\mathbf{A}} - \mathbf{C}\|_{\square} \leq \epsilon$$

$\mathbf{C} = \mathbf{Q}\mathrm{diag}(\mathbf{r})\mathbf{Q}^{\top}$ and has $K = \frac{-9\log(\epsilon/2)^2}{\epsilon^2}$ communities.

PieClam Is Universal

Let $\mathbf{r}_+ = \mathrm{ReLU}(\mathbf{r})$ and $\mathbf{r}_- = \mathrm{ReLU}(-\mathbf{r})$. Define:

$$\mathbf{T} = \mathbf{Q}\mathrm{diag}(\sqrt{\mathbf{r}_+})$$

$$\mathbf{S} = \mathbf{Q}\mathrm{diag}(\sqrt{\mathbf{r}_-})$$

$$\mathbf{F} = \left(\mathbf{T}, \mathbf{S}\right)$$

It is easy to see that $\mathbf{F}^\top \mathbf{L} \mathbf{F} = \mathbf{C}$

Define the Clam matrix $\mathbf{P} = \left(1 - \exp(\mathbf{f}_n^\top \mathbf{L} \mathbf{f}_m)\right)_{n,m \in [N] \times [N]}$

PieClam Is Universal

Substituting $\mathbf{F}^\top\mathbf{L}\mathbf{F}$ for $\mathbf{C}$ we get that

$$\|\tilde{\mathbf{A}} - \mathbf{F}^\top\mathbf{L}\mathbf{F}\|_\square \leq \epsilon$$

It is also easy to see that

$$\|\tilde{\mathbf{A}} - \mathbf{F}^\top\mathbf{L}\mathbf{F}\|_\square =$$

$$\frac{1}{N^2} \sup_{\mathcal{U},\mathcal{V}\subset[N]} \left| \sum_{n\in\mathcal{U}} \sum_{m\in\mathcal{V}} \Big(-\log\left(1-(1-\epsilon)a_{n,m}\right) - \log\left(1-p_{n,m}\right)\Big) \right|$$

So by definition of the log cut as the infimum (substituting $d = \epsilon$):

$$D(\mathbf{A}\|\mathbf{P}) \leq \|\tilde{\mathbf{A}} - \mathbf{F}^\top\mathbf{L}\mathbf{F}\|_\square + \epsilon$$

Finally:

$$D(\mathbf{A}\|\mathbf{P}) \leq 2\epsilon$$

Q.E.D.

Universality of PieClam in $\mathcal{T}$

For this result, we use the standard weak regularity lemma for non-intersecting classes.

Definition

A *block matrix* $\mathbf{B}$ with K classes is a symmetric matrix $\mathbf{B} \in [0,\infty)^{N\times N}$ for which there exists a partition of $[N]$ into K disjoint sets, called *classes*, $\mathcal{C}_1,\ldots,\mathcal{C}_K$ (with $\cup\mathcal{C}_j = [N]$), such that for every pair of classes $i,j \in [K]$, there is a constant $c_{i,j} \geq 0$ such that $b_{n,m} = c_{i,j}$ for any two nodes $n \in \mathcal{C}_i$ and $m \in \mathcal{C}_j$.

Theorem

Let $\mathbf{A} \in [0,R]^{N\times N}$ *be an adjacency matrix of a graph with* N *nodes. Let* $\epsilon > 0$*. Denote* $K = 2^{2\lceil R^2/\epsilon^2\rceil}$*. Then, there exists a block matrix* $\mathbf{B}$ *with* K *(disjoint) classes such that*

$$\|\mathbf{A}-\mathbf{B}\|_{\square} \leq \epsilon. \tag{6}$$

Universality of PieClam in $\mathcal{T}$

Let $\epsilon > 0$. Let $\mathbf{A} \in [0,1]^{N \times N}$ be an adjacency matrix.
Consider the matrix $\tilde{\mathbf{A}} \in [0, -\log(\epsilon/2)]$ with entries

$$\tilde{a}_{n,m} = -\log(1 - (1-\epsilon/2)a_{n,m}).$$

Goal: we build IeClam affiliation features $\mathbf{F}$ such that

$$1 - \exp(-\mathbf{f}_n^\top \mathbf{L} \mathbf{f}_m) \approx (1-\epsilon/2)a_{n,m}.$$

Universality of PieClam in $\mathcal{T}$

- By the weak regularity lemma:

$$\|\tilde{\mathbf{A}} - \mathbf{B}\|_{\square} \leq \epsilon.$$

  $\mathbf{B}$ is a block matrix with $K = 2^{2\lceil -\log(\epsilon/2)^2/\epsilon^2 \rceil}$ classes $C_1, \ldots, C_K$

- For each pair of classes $i, j \in [K]$, let $c_{i,j} \geq 0$ denote the edge weight between C_i and C_j.

- We Build $\mathbf{B}$ using K^2 inclusive and K^2 exclusive communities so $\mathbf{B} = \mathbf{F}^\top L \mathbf{F}$

Construction

- For every $n \in [N]$, Define k_n s.t. $n \in C_{k_n}$
- Set:
  $t_n^c = \sqrt{c_{k_n,k_n}}$ at channel $c = (K-1)k_n + k_n$
  $(t_n^k, s_n^k) = \left(\frac{\sqrt{c_{k_n,j}}}{4}, -\frac{\sqrt{c_{k_n,j}}}{4}\right)$ for $k = (k_n - 1)K + j$ and $k = k_n + (j-1)K$
- In all other channels t_n^k and s_n^k are zero.
- Note that $\mathbf{f}_n$ belongs to the cone of pairwise non-negativity $\mathcal{T} \subset \mathbb{R}^{2K^2}$.
- It is now direct to see that $\mathbf{f}_n^\top L \mathbf{f}_m = c_{k_n,k_m} = b_{nm}$.

As a result

$$D_{\square}(\mathbf{P}||\mathbf{A}) \leq$$

$$\epsilon/2 + \frac{1}{N^2} \sup_{\mathcal{U},\mathcal{V}\subset[N]} \left| \log\left(\prod_{n\in\mathcal{U}} \prod_{m\in\mathcal{V}} \frac{1-p_{n,m}}{1-(1-\epsilon/2)a_{n,m}} \right) \right|$$

$$= \epsilon/2 + \frac{1}{N^2} \sup_{\mathcal{U},\mathcal{V}\subset[N]} \left| \sum_{n\in\mathcal{U}} \sum_{m\in\mathcal{V}} \Big(-\log\left(1-(1-\epsilon)a_{n,m}\right) + \log\left(1-p_{n,m}\right) \Big) \right|$$

$$= \epsilon/2 + \|\tilde{\mathbf{A}} - \mathbf{B}\|_{\square} = \epsilon.$$